

When Pruning Meets Interpretability: Preserving Sparse Autoencoder Robustness in LLMs

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Abstract

Sparse autoencoders (SAEs) are widely used to interpret the internal representations of large language models (LLMs), yet their reliability under post-hoc model compression remains poorly understood. We present a systematic study of how pruning affects SAE behavior and theoretically show that, for a fixed SAE, its impact is governed by perturbation energy, a covariance-weighted norm. This perspective exposes a key limitation of MAGNITUDE pruning: by ignoring activation geometry, it distorts the learned representation space and degrades SAE functionality. Activation-aware methods such as WANDA and SPARSEGPT, in contrast, implicitly control perturbation energy and are therefore substantially more robust at preserving SAE behavior. We further reveal a consistent structural vulnerability across all pruning methods: middle layers are significantly more sensitive to pruning than early or late layers. Guided by this insight, we propose a Layer-wise sparsity allocation strategy, achieving lower perplexity under the same average pruning sparsity. Experiments across four model architectures validate our theoretical findings. The code is publicly available at [sae-robustness-under-pruning](#).

1 Introduction

The superposition hypothesis (Elhage et al., 2022) posits that neural networks represent more features than they have neurons, giving rise to polysemanticity where individual neurons respond to multiple unrelated concepts. Sparse autoencoders (SAEs) (Cunningham et al., 2023) have emerged as the leading method for resolving this: by training an overcomplete sparse dictionary on a model’s activations, they recover monosemantic, interpretable features (Bricken et al., 2023; Zhu et al., 2026), an approach that scales to frontier models (Templeton et al., 2024; Lieberum et al., 2024). SAE-based analysis has consequently become a core workflow in mechanistic interpretability (Shu et al., 2025; Sharkey et al., 2025). SAEs are used to identify circuits (Conmy et al., 2023; Wang et al., 2023; Chanin et al., 2025), erase spurious correlations (Karvonen et al., 2024), and test causal hypotheses about model behavior. All of these applications rely on the implicit assumption that any SAE provides a faithful and reliable decomposition of the underlying activations to which it is applied.

Independently, weight pruning has become a standard post-hoc technique for reducing the computational cost of deploying large language models. Methods ranging from classical MAGNITUDE pruning (Han et al., 2016) to modern one-shot approaches such as SPARSEGPT (Frantar & Alistarh, 2023) and WANDA (Sun et al., 2024) can remove a substantial fraction of model weights with only modest degradation in downstream task performance. Given the prohibitive cost of deploying large dense models, compression techniques including pruning, quantization, and distillation are standard tools in the open-weight deployment pipeline (Muralidharan et al., 2024; Ma et al., 2023; Guo et al., 2025; Cai et al., 2024; Zhu et al., 2025), often applied after interpretability analysis has been conducted on the dense checkpoint and any associated SAEs have been trained. Crucially, the criteria by which these methods select weights to prune: whether by magnitude alone, by activation

statistics, or by second-order Hessian, differ substantially in how they perturb the model’s internal activation distribution, a distinction that has received little attention outside of predictive performance evaluations.

These two workflows are inherently in tension. An SAE is trained on the activation distribution of a specific dense model, whereas pruning alters that model’s weight matrices, perturbing all downstream activations. When these perturbations remain small relative to the SAE’s training distribution, the SAE may continue to function approximately as intended. Otherwise, its latents can fail silently (Chanin et al., 2025; Karvonen et al., 2024; Gupte et al., 2025), and importantly, such failures may not be detected by standard language modeling benchmarks (Egashira et al., 2026; Sharkey et al., 2025). Despite this, there is currently no theoretical framework that characterizes when pruning preserves SAE validity, nor an explanation for why different pruning methods lead to varying degrees of degradation.

A natural response to a changed model is to retrain the interpretability tooling itself. Cross-coders (Lindsey, 2024), for instance, train a joint dictionary across two model states (e.g., base and fine-tuned) to discover what changed between them. This is effective, but it requires training a new dictionary for each pair of model variants, precisely the retraining cost we aim to avoid. Our work instead asks: *given an SAE already trained on a dense model, under what conditions does it remain valid after pruning, without retraining?* This is the more common deployment scenario, since organizations that have invested resources in SAE training need to know whether existing analysis transfers to a compressed model rather than needing to characterize what changed. The two approaches are thus complementary: crosscoders are the right tool for understanding a specific change, whereas answering the transfer question requires a theoretical account of when pruning leaves an existing SAE intact, which is exactly what is missing. This paper provides that account.

We begin by theoretically examining the impact of MAGNITUDE, SPARSEGPT, and WANDA pruning, and show that SAE degradation is governed by **perturbation energy** (ϵ^2), a covariance-weighted norm that weights perturbations more heavily along high-variance input directions. This perspective establishes a formal separation among the three pruning methods: each method can be seen as minimizing a mathematically distinct approximation of ϵ^2 . MAGNITUDE pruning, which ignores activation geometry, induces the largest perturbation energy and the most severe SAE degradation, whereas activation-aware methods such as WANDA and SPARSEGPT implicitly control this quantity and are substantially more robust at preserving SAE behavior.

We validate our theory using four SAEBench (Karvonen et al., 2025) metric categories: Core (Karvonen et al., 2025), Feature Absorption (Chanin et al., 2025), SCR, and TPP (Karvonen et al., 2024), which together capture the full range of qualitative SAE failure modes. Beyond comparing methods, our experiments reveal a consistent structural vulnerability: **middle layers are significantly more sensitive to pruning than early or late layers**, across all methods and sparsity levels. Building on this insight, we explore a **layer-wise sparsity allocation strategy** that preserves SAE quality and achieves lower perplexity while maintaining the same average sparsity level.

The remainder of the paper is organized as follows. Sec. 2 presents background on sparse autoencoders and formulates pruning as activation perturbation. Sec. 3 develops the perturbation-theoretic framework and derives the formal separation among pruning methods. Sec. 4 argues for the sufficiency of the SAEBench evaluation suite in capturing all relevant SAE failure modes. Sec. 5 describes the experimental setup, and Sec. 6 presents the results. Sec. 7 concludes the paper and discusses limitations and future directions.

2 Preliminaries

2.1 Sparse autoencoders for mechanistic interpretability

Definition 2.1 (Sparse Autoencoder). A sparse autoencoder (Cunningham et al., 2023) is a map $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$ with encoder $E_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^m$ and decoder $D_\theta : \mathbb{R}^m \rightarrow \mathbb{R}^d$, where $m \gg d$:

$$f_\theta(x) = D_\theta(\sigma(E_\theta(x))), \quad E_\theta(x) = W_{\text{enc}}x + b_{\text{enc}}, \quad D_\theta(z) = W_{\text{dec}}z + b_{\text{dec}}, \quad (1)$$

where σ is an arbitrary activation function, $W_{\text{enc}} \in \mathbb{R}^{m \times d}$, $W_{\text{dec}} \in \mathbb{R}^{d \times m}$.

The SAE is trained on activations $x \sim \mathcal{D}$ from the model’s forward pass to minimize:

$$\mathcal{L}_{\text{SAE}} = \mathbb{E}_{x \sim \mathcal{D}} \left[\|x - f_{\theta}(x)\|^2 + \lambda \|\sigma(E_{\theta}(x))\|_1 \right]. \quad (2)$$

A key property we assume is that trained SAEs are Lipschitz continuous.

Assumption 2.2 (Lipschitz continuity of the SAE). The trained SAE f_{θ} is L -Lipschitz:

$$\|f_{\theta}(x') - f_{\theta}(x)\| \leq L \|x' - x\| \quad \forall x, x' \in \mathbb{R}^d. \quad (3)$$

This assumption holds for standard SAE architectures whenever coordinate-wise activation function σ is Lipschitz, as satisfied by common choices such as ReLU and GELU.

2.2 Weight pruning as activation perturbation

Consider a linear layer with weight matrix $W \in \mathbb{R}^{d \times p}$ producing activation $x = Wx_{\text{in}}$, where $x_{\text{in}} \in \mathbb{R}^p$ is the layer input. Pruning modifies the weight matrix by zeroing a subset of entries $S \subset [d] \times [p]$, resulting in $W' = W + \Delta W$. The corresponding pruned activation is

$$x' = W'x_{\text{in}} = x + \delta, \quad \delta := \Delta W x_{\text{in}}. \quad (4)$$

We next summarize the objectives of common pruning methods.

Definition 2.3 (Pruning objectives). Let S be the pruning mask at target sparsity s .

1. **MAGNITUDE pruning** (Han et al., 2016) selects S to minimize $\|\Delta W\|_F^2 = \sum_{(i,j) \in S} w_{ij}^2$.
2. **WANDA** (Sun et al., 2024) selects S to minimize $\sum_{(i,j) \in S} w_{ij}^2 \cdot \mathbb{E} \left[(x_{\text{in}})_j^2 \right]$.
3. **SPARSEGPT** (Frantar & Alistarh, 2023) solves layer-wise regression to approximately minimize $\mathbb{E} [\|\Delta W x_{\text{in}}\|^2]$ via second-order optimization.

3 Perturbation Theory: SAE Degradation under Pruning

Next, we theoretically examine how pruning affects SAE behavior. Specifically, we develop a unified perturbation-theoretic analysis that bounds SAE degradation in terms of a data-dependent quantity, which we call the perturbation energy, together with the SAE’s Lipschitz constant. This quantity provides a theoretical foundation for understanding the varying impact of different pruning techniques and explains why SAEs can be robust to some methods while vulnerable to others. Proof details are provided in Appendix A.1.

Impact of pruning on SAE. Recall from Sec. 2.2 that pruning replaces W with $W' = W + \Delta W$, shifting the activation by $\delta = \Delta W x_{\text{in}}$. Because the SAE f_{θ} is L -Lipschitz (Assumption 2.2), the triangle inequality immediately gives a per-sample reconstruction bound.

Lemma 3.1. Let f_{θ} be L -Lipschitz and let $x' = x + \delta$. Then:

$$\|x' - f_{\theta}(x')\| \leq \|x - f_{\theta}(x)\| + (1 + L)\|\delta\|. \quad (5)$$

Squaring, taking expectations, and applying Cauchy–Schwarz to the cross term yields Thm. 3.2, which bounds expected degradation in SAE reconstruction quality after pruning.

Theorem 3.2 (Expected reconstruction degradation). Under Assumption 2.2,

$$\mathbb{E} [\|x' - f_{\theta}(x')\|^2] \leq \underbrace{\mathbb{E} [\|x - f_{\theta}(x)\|^2]}_{\text{intrinsic SAE error}} + (1 + L)^2 \varepsilon^2 + 2(1 + L) \sqrt{\mathbb{E} [\|x - f_{\theta}(x)\|^2] \cdot \varepsilon^2} \quad (6)$$

where the **perturbation energy** ε^2 admits the covariance decomposition

$$\varepsilon^2 := \mathbb{E} [\|\delta\|^2] = \mathbb{E} [\|\Delta W x_{\text{in}}\|^2] = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^{\top}), \quad (7)$$

with $\Sigma_{x_{\text{in}}} = \mathbb{E} [x_{\text{in}} x_{\text{in}}^{\top}]$ the second-moment matrix of layer inputs.

Although Thm. 3.2 is an upper bound, it provides insight into what drives SAE reconstruction degradation after pruning. The bound decomposes into two components: (1) *intrinsic SAE error* which reflects how well the SAE was trained on the original model and is independent of pruning; (2) *perturbation energy* ε^2 which captures the direct damage caused by pruning. The third term captures their interaction, showing that a poorly trained SAE amplifies the effect of pruning: even small ε^2 can cause disproportionately large degradation when the intrinsic error is large.

The bound also depends on the Lipschitz constant L , a property of the SAE architecture and training that is independent of the pruning method: both the $(1 + L)^2$ and cross terms grow with L , so a less smooth SAE amplifies any given perturbation energy. Since we hold SAE weights fixed throughout all experiments, L is constant across pruning methods, making ε^2 the only operative variable when comparing methods; all method-level statements in this paper should be read under this fixed-SAE assumption.

If we write the eigen decomposition $\Sigma_{x_{\text{in}}} = \sum_i \lambda_i v_i v_i^\top$ with $\lambda_1 \geq \dots \geq \lambda_p > 0$, then

$$\varepsilon^2 = \sum_i \lambda_i \|\Delta W v_i\|^2. \quad (8)$$

This is a *covariance-weighted* squared norm of ΔW : each term $\lambda_i \|\Delta W v_i\|^2$ quantifies how strongly ΔW perturbs the i -th principal direction of the input distribution, scaled by the variance along that direction. Perturbations aligned with high-variance directions contribute most to ε^2 , while those along low-variance directions have little effect. The impact of pruning on SAEs therefore depends not only on how much the weights change, but on how those changes align with the input covariance structure, and pruning methods differ precisely in how well they account for this geometry.

Separation among pruning methods. The covariance weighting in (8) induces a fundamental distinction between MAGNITUDE and activation-aware pruning methods.

Corollary 3.3 (Comparison of pruning methods). *At any target sparsity s :*

- **MAGNITUDE pruning** minimizes $\|\Delta W\|_F^2 = \text{tr}(\Delta W I \Delta W^\top)$, which coincides with minimizing ε^2 only when $\Sigma_{x_{\text{in}}} \propto I$.
- **WANDA** minimizes $\text{tr}(\Delta W \text{diag}(\mathbb{E}[x_{\text{in}} x_{\text{in}}^\top]) \Delta W^\top)$, a diagonal approximation to ε^2 .
- **SPARSEGPT** approximately minimizes ε^2 directly via second-order information.

Corollary 3.3 highlights that the three pruning methods differ fundamentally in how they approximate ε^2 : MAGNITUDE pruning, by ignoring the covariance structure of the activations, approximates ε^2 poorly and consequently induces larger perturbation energy and more severe SAE degradation. In contrast, WANDA and SPARSEGPT implicitly account for this structure, better controlling ε^2 and thus more effectively preserving SAE behavior.

This separation can be made precise: the three methods form a natural hierarchy in how faithfully they approximate the true objective ε^2 :

$$\underbrace{\text{tr}(\Delta W I \Delta W^\top)}_{\text{MAGNITUDE}} \prec \underbrace{\text{tr}(\Delta W \text{diag}(\Sigma_{x_{\text{in}}}) \Delta W^\top)}_{\text{WANDA}} \prec \underbrace{\text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)}_{\text{SparseGPT}}, \quad (9)$$

where $\prec$ denotes “is a coarser approximation of the true perturbation energy than”. MAGNITUDE pruning ignores $\Sigma_{x_{\text{in}}}$ entirely; WANDA uses only its diagonal; SparseGPT uses the full matrix, mirroring exactly the ordering of SAE degradation we observe empirically in Sec 6.

4 From Theory to Evaluation: SAEBench as a Sufficient Validation Suite

Sec. 3 shows that, for a fixed SAE, ε^2 governs SAE degradation under pruning. Validating this empirically requires metrics that collectively cover every way an SAE can fail when its input distribution shifts. We argue that the four SAEBench categories, *Core*, *Feature Absorption*, *SCR*, and *TPP*, are jointly sufficient as they cover three orthogonal failure modes.

Omitting any one would leave a blind spot in the validation. We describe each failure mode and its corresponding metric in turn, then argue their joint sufficiency (see Appendix A.2 for metric definitions).

Failure Mode 1: Reconstruction Degrades. When pruning shifts the activation distribution, the SAE may no longer accurately reconstruct its inputs. Since the SAE’s reconstruction is inserted back into the model’s forward pass, degraded reconstruction directly corrupts the model’s predictions. The **Core** metrics (Karvonen et al., 2025) detect this failure by measuring how well the SAE’s reconstruction preserves model behavior, using the KL divergence score, CE loss score, explained variance, MSE, and cosine similarity (see Appendix A.2.1). Preserved Core metrics indicate that the SAE remains within its effective reconstruction region after pruning. However, Core metrics are aggregated over all latents and tokens, making them insensitive to localized failures: a score of 0.99 is consistent with 5% of latents behaving incorrectly, provided the rest compensate. The SAE can silently reorganize its internal representations while keeping aggregate scores high, leaving the failure modes below entirely undetected.

Failure Mode 2: Features Break. Even when aggregate reconstruction appears intact, individual latents may fail. A latent that should activate on a given input may go silent (false negative), or one that should be silent may fire spuriously (false positive). When this happens, the SAE no longer provides a faithful, interpretable decomposition of the activations, even though nothing appears wrong at the aggregate level. **Feature Absorption** (Chanin et al., 2025) detects this failure by testing whether the SAE maintains clean feature separation after pruning. Using a first-letter classification task, it identifies cases where an expected latent fails to activate and checks whether a semantically unrelated latent has absorbed its information. A lower absorption rate indicates better feature separation. Thus, the absorption metric goes beneath the aggregate surface and checks whether a feature that should represent “starts with S” still fires on S-tokens, or whether its information has been absorbed by a semantically unrelated latent. Without this metric, we could observe perfect Core scores while the SAE has been silently rearranged into an uninterpretable state.

Failure Mode 3: Interventions Stop Working. The primary use of SAEs in interpretability is causal intervention: ablating specific latents to test whether they causally mediate a concept or behavior. Pruning can break this connection: a latent may still activate on the right inputs but no longer causally influence the model’s output as expected, or its ablation may collaterally affect unrelated concepts. **Spurious Correlation Removal (SCR)** (Karvonen et al., 2024) detects this failure by testing whether latent ablations still produce their intended causal effect after pruning: if ablating latents S_c removed a spurious “gender $\rightarrow$ profession” correlation in the dense model, SCR checks whether the same ablation achieves the same effect after pruning. Without SCR, we could observe latents that activate correctly yet have lost their causal influence: the latent fires on the right inputs, but ablating it no longer changes the model’s behavior. **Targeted Probe Perturbation (TPP)** (Karvonen et al., 2024) complements SCR by testing *selectivity*: ablating latents for concept c should degrade performance on class c but leave other classes unaffected. Without TPP, we could conclude that interventions remain effective while missing that they have become non-specific, damaging concepts they were never intended to affect.

Joint Sufficiency. Together, these metrics form a logical chain that mirrors the structure of our perturbation theory. **Core** verifies that ϵ^2 is small enough for aggregate reconstruction to be preserved (Thm. 3.2). **Feature Absorption** verifies that ϵ^2 is small enough relative to activation margins, ensuring individual latents remain intact. **SCR** verifies that surviving features retain their causal role, so that latent-level interventions still produce the intended effect. **TPP** verifies that these causal effects remain selective, that concepts stay disentangled, not merely detectable. This progression from aggregate fidelity, to feature-level correctness, to interventional validity, to interventional selectivity covers the full stack of properties that make SAEs useful as an interpretability tool. If all four are preserved under pruning, every standard SAE-based analysis transfers from the dense model to the pruned model.

5 Experimental Setup

Models and SAEs. To isolate the effect of pruning from SAE training variability, we hold all SAE weights fixed and use pretrained checkpoints from SAELens (Bloom et al., 2024). Our model selection is governed by the availability of high-quality pretrained SAEs. By fixing pretrained SAEs, we ensure that any observed degradation is attributable solely to pruning and not to SAE training choices. This consideration leads us to four model–SAE pairs spanning roughly two orders of magnitude in parameter count: pythia-70m (Biderman et al., 2023; Jedryszek & Crook, 2026), gemma-2-2b (Team et al., 2024; Lieberum et al., 2024), gemma-2-9b (Team et al., 2024; Lieberum et al., 2024), and mistral-7b (Jiang et al., 2023; Engels, 2024), all of which have publicly available, well-validated SAE checkpoints. Our main experiments focus on gemma-2-2b, which benefits from the most comprehensive SAE coverage among the four models. This model has pretrained SAEs available at every residual-stream layer (layers 0–25), enabling the fine-grained layer-level analysis that is central to our investigation. Results for all three remaining models are reported in Sec. 6 as a cross-model generalisation study, where we verify that the theoretical ordering and layer sensitivity patterns observed on gemma-2-2b hold across architectures and scales.

Pruning methods. We apply three pruning methods post-hoc to each dense checkpoint: MAGNITUDE pruning (Han et al., 2016), WANDA (Sun et al., 2024), and SPARSEGPT (Frantar & Alistarh, 2023) (see Sec. 2.2 for an overview). No retraining or weight updates are performed after pruning. Calibration data for WANDA and SPARSEGPT consists of 128 samples from OpenWebText (Gokaslan et al., 2019). We conduct experiments at 50% sparsity, a practically relevant operating point at which all three methods remain functional and differences in SAE degradation are clearly visible. To characterise how SAE degradation scales with the degree of compression, we additionally evaluate at 25% and 40% sparsity, allowing us to assess whether the layer sensitivity patterns and method ordering identified at 50% are consistent across sparsity levels or specific to that operating point.

Evaluation. For each pruned model variant, we compute the four SAE Bench (Karvonen et al., 2025) metric categories described in Sec. 4: Core (KL divergence score, CE loss score, explained variance, cosine similarity, MSE), Feature Absorption, SCR, and TPP. Evaluations are run on the residual output of every layer (layers 0–25 for gemma-2-2b) to expose layer-level sensitivity patterns. For SCR and TPP, we adopt $k = 10$ as the canonical ablation threshold, as it lies in the stable regime of the threshold sensitivity curve and produces the clearest separation between pruning methods (see Appendix A.2). All results are reported both as raw metric values and as percentage change from the dense baseline.

6 Results

Impact of pruning methods is consistent and theory-aligned. Table 1 and Figure 1 together reveal a clear and consistent ordering across all SAE Bench metric categories: MAGNITUDE pruning causes the most severe SAE degradation, WANDA is substantially more robust, and SPARSEGPT is equally conservative, closely tracking the dense baseline. This ordering holds without exception across all eight metrics and all 26 layers of gemma-2-2b at 50% sparsity.

The pattern precisely validates our perturbation-theoretic analysis: MAGNITUDE pruning minimises the wrong objective: the Frobenius norm, which equals perturbation energy only under an identity covariance. While WANDA and SPARSEGPT progressively better approximate the true perturbation energy ϵ^2 . The separation is particularly stark on interventional metrics: MAGNITUDE pruning degrades SCR by up to 60% and TPP by up to 80% relative to the dense baseline in the worst-affected layers, whereas WANDA and SPARSEGPT remain within 15–25% of baseline on the same metrics (Refer to Figure 1).

Critically, the Core metrics, which aggregate over all latents, substantially understate the damage inflicted by MAGNITUDE pruning. For instance, KL divergence scores for MAGNITUDE pruning in middle layers appear superficially reasonable (≥ 0.96), yet the corresponding SCR and TPP scores collapse, confirming our theoretical prediction that aggregate reconstruction fidelity is insufficient to certify SAE reliability. Appendix A.4 closes the loop

Layers	Method	Spar. (%)	Core					Absorption		SCR	TPP
			KL Score↑	CE Score↑	Expl. Var.↑	Cos Sim↑	MSE ↓	Abs. Frac.↓	Full Abs.↓	Top-10 ↑	Top-10 ↑
Early (0–8)	<i>Baseline</i>	0	0.9944	0.9941	0.8976	0.9497	0.4268	0.0674	0.0354	0.1714	0.0270
	MAGNITUDE	50	0.9847	0.9818	0.8416	0.9175	0.6536	0.0764	0.0486	0.1130	0.0187
	WANDA	50	0.9935	0.9873	0.8793	0.9401	0.4648	0.0691	0.0353	0.1487	0.0188
	SPARSEGPT	50	0.9918	0.9906	0.8728	0.9353	0.4886	0.0717	0.0402	0.1329	0.0196
Middle (9–17)	<i>Baseline</i>	0	0.9887	0.9876	0.8763	0.9301	2.0582	0.0668	0.0586	0.3047	0.0424
	MAGNITUDE	50	0.9644	0.9531	0.7782	0.8780	3.3021	0.0794	0.0730	0.1301	0.0089
	WANDA	50	0.9894	0.9854	0.8507	0.9175	2.2214	0.0660	0.0564	0.2399	0.0304
	SPARSEGPT	50	0.9868	0.9893	0.8277	0.9071	2.5616	0.0731	0.0662	0.2103	0.0258
Late (18–25)	<i>Baseline</i>	0	0.9776	0.9765	0.8682	0.9307	14.2891	0.1208	0.1369	0.3885	0.0681
	MAGNITUDE	50	0.9535	0.9510	0.7544	0.8701	29.4766	0.0831	0.1177	0.3192	0.0117
	WANDA	50	0.9780	0.9773	0.8569	0.9253	14.7422	0.1045	0.1179	0.3743	0.0519
	SPARSEGPT	50	0.9768	0.9815	0.8389	0.9150	17.6016	0.1054	0.1293	0.3605	0.0547

Table 1: SAEBench metrics at 50% sparsity for gemma-2-2b, aggregated by layer group (Early: layers 0–8; Middle: 9–17; Late: 18–25). Each group reports the unpruned dense baseline (italicised) alongside all three pruning methods. ↑ higher is better; ↓ lower is better. Refer to App. A.3 for per-layer SAEBench metrics from which these values are aggregated.

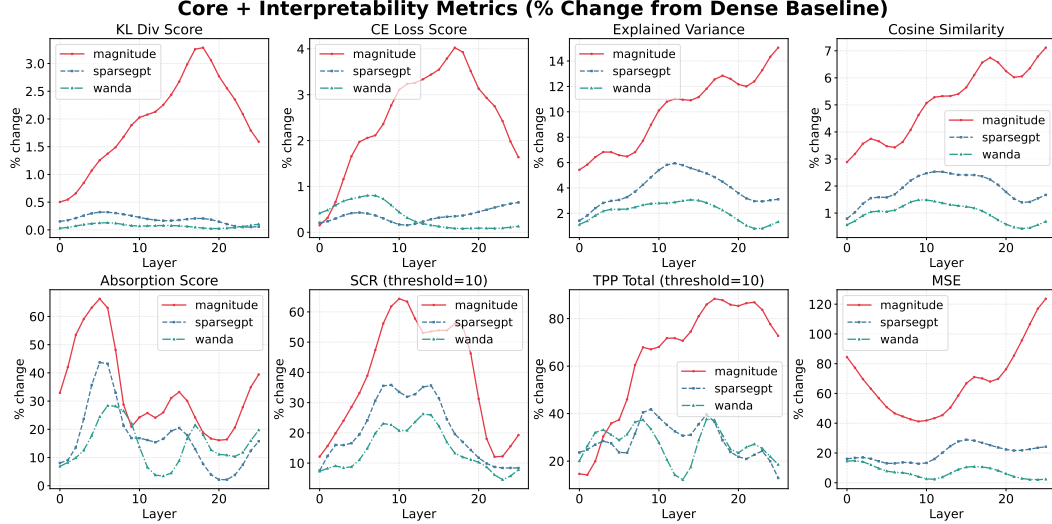


Figure 1: Per-layer percentage degradation from the dense baseline across all eight SAEBench metric categories for gemma-2-2b at 50% sparsity. Each panel corresponds to one metric; MAGNITUDE (red), SPARSEGPT (blue), and WANDA (green) across layers 0–25. These plots emphasize per-layer trends and direct cross-method comparison at each layer; see Figure 2 for the same data as a heatmap.

directly: measured ϵ^2 values confirm the ordering of Corollary 3.3 at every layer group, and reveal that middle layers have the *lowest* local ϵ^2 despite the highest degradation, implicating accumulated upstream perturbation.

Middle layers exhibit systematic structural vulnerability. A striking and consistent finding across all methods and metrics is the elevated sensitivity of middle layers (9–17) to pruning, relative to both early layers (0–8) and late layers (18–25). Figure 3 makes this explicit: averaging the percentage degradation across all methods and metrics, the middle-layer profile rises sharply above that of early and late layers at every sparsity level tested (25%, 40%, 50%), and this pattern is monotone in sparsity, confirming it is not an artifact of a particular operating point. At 50% sparsity, the heatmap in Figure 2 shows that middle layers

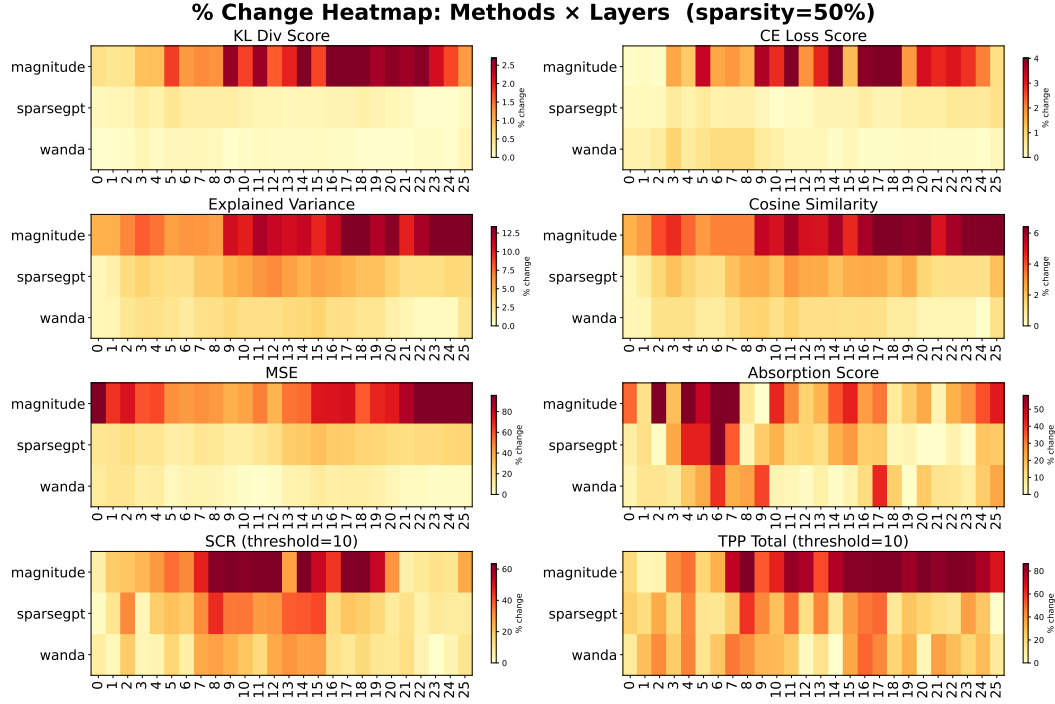


Figure 2: Heatmap of per-layer, per-method percentage degradation from the dense baseline for gemma-2-2b at 50% sparsity. Each row corresponds to a pruning method; each column to a layer; colour intensity encodes the magnitude of degradation (darker = larger change). This heatmap presents the same underlying data as Figure 1, reorganized to expose spatial patterns across the full method $\times$ layer grid simultaneously.

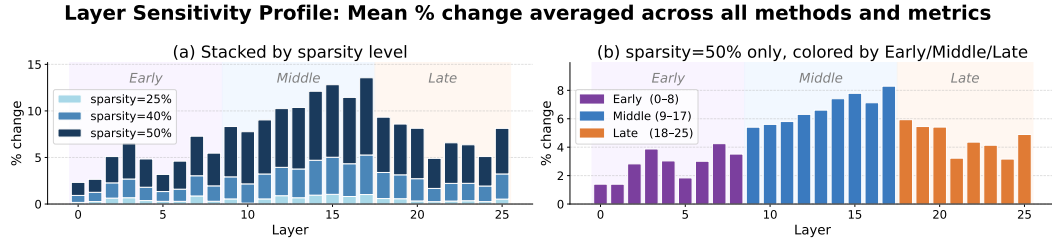


Figure 3: Layer sensitivity profile for gemma-2-2b, computed as the mean absolute percentage degradation averaged across all three pruning methods and all eight SAEbench metrics. **(a)** Sensitivity profiles at three sparsity levels (25%, 40%, 50%). **(b)** Sensitivity profile at 50% sparsity, coloured by layer group (Early: 0–8, Middle: 9–17, Late: 18–25).

carry the darkest cells across nearly every metric–method combination, with MAGNITUDE pruning on middle layers producing degradations of 10–14% in explained variance, 5–7% in cosine similarity, and 40–60% in SCR and TPP, substantially exceeding the corresponding values for early and late layers.

We hypothesize this reflects the asymmetric propagation of representational damage across depth. Early layers construct foundational features that are heavily inherited by middle layers via residual connections, making middle layers the primary recipients of perturbation energy ε^2 introduced by pruning upstream weights. Late layers, by contrast, operate on an already fully composed representation and specialize in output-facing transformations, leaving them with little residual upstream dependency to corrupt. This asymmetry directly motivates a layer-wise sparsity schedule.

Model	Method	Spar. (%)	Core					Absorption		SCR	TPP
			KL Score $\uparrow$	CE Score $\uparrow$	Expl. Var. $\uparrow$	Cos Sim $\uparrow$	MSE $\downarrow$	Abs. Frac. $\downarrow$	Full Abs. $\downarrow$	Top-10 $\uparrow$	Top-10 $\uparrow$
gemma-2 2b	Baseline	0	0.9873	0.9864	0.8812	0.9370	5.2568	0.0836	0.0746	0.2843	0.0450
	MAGNITUDE	50	0.9681	0.9624	0.7928	0.8893	10.4390	0.0795	0.0783	0.1824	0.0131
	WANDA	50	0.9873	0.9835	0.8625	0.9277	5.4659	0.0789	0.0680	0.2497	0.0330
	SPARSEGPT	50	0.9855	0.9874	0.8468	0.9193	6.4717	0.0826	0.0766	0.2297	0.0325
gemma-2 9b	Baseline	0	0.9914	0.9900	0.8882	0.9380	7.7362	0.0997	0.0654	0.3237	0.0603
	MAGNITUDE	50	0.9826	0.9757	0.8506	0.9173	12.4004	0.0919	0.0619	0.2307	0.0298
	WANDA	50	0.9911	0.9869	0.8772	0.9322	7.7391	0.0905	0.0608	0.2921	0.0409
	SPARSEGPT	50	0.9901	0.9892	0.8627	0.9245	9.6890	0.0910	0.0606	0.2706	0.0395
pythia 70m	Baseline	0	0.9262	0.9240	0.9476	0.9572	0.0118	N/A ⁺	N/A ⁺	0.0726	0.0505
	MAGNITUDE	50	0.6422	0.8740	0.5592	0.8765	0.0615	N/A ⁺	N/A ⁺	0.0054	0.0420
	WANDA	50	0.8798	0.9091	0.8705	0.9244	0.0195	N/A ⁺	N/A ⁺	0.0150	0.0405
	SPARSEGPT	50	0.8819	0.9198	0.9025	0.9328	0.0204	N/A ⁺	N/A ⁺	0.0321	0.0409
mistral 7b	Baseline	0	0.9722	0.9753	0.9844	0.8490	0.0150	0.4292	0.4150	0.3627	0.0501
	MAGNITUDE	50	0.9672	0.9758	0.9792	0.8034	0.0177	0.4510	0.4371	0.3634	0.0208
	WANDA	50	0.9742	0.9720	0.9857	0.8477	0.0128	0.4023	0.3805	0.3810	0.0463
	SPARSEGPT	50	0.9749	0.9734	0.9870	0.8477	0.0131	0.3848	0.3715	0.3617	0.0396

⁺ Feature absorption evaluation requires sufficient first-letter features to be detectable in the model. SAEbench (Karvonen et al., 2025) issues a warning that this evaluation is only reliable for models ≥ 2 B parameters; pythia-70m (Biderman et al., 2023) (70M parameters) does not produce enough first-letter features for a valid absorption measurement (Chanin et al., 2025; Jedryszek & Crook, 2026), so these cells are excluded from all absorption analyses and averages.

[†] Pretrained SAEs for mistral-7b are available for only one layer per depth group (early, middle, late) in SAELens (Bloom et al., 2024; Engels, 2024); mistral-7b values are therefore averaged over 3 layers. The method ordering is consistent with the full-coverage models, but cross-model magnitude comparisons involving mistral-7b should be interpreted with this coverage difference in mind.

Table 2: Cross-architecture generalisation at 50% sparsity. Each cell reports the metric value averaged across all evaluated layers of a given model. $\uparrow$ higher is better; $\downarrow$ lower is better.

Layer-wise sparsity allocation. The layer sensitivity profile identified earlier suggests that a uniform sparsity schedule is suboptimal. Motivated by this hypothesis, we experiment with redistributing the global compression budget unevenly across depth: *assigning lower sparsity to early layers to preserve the activations that middle layers depend on, and higher sparsity to late layers where upstream perturbation has a negligible downstream consequence*. Concretely, given a target global sparsity $\bar{s}$, we assign layer-wise sparsity s_ℓ according to a schedule that ramps linearly from $s_{\min}$ at the first layer to a plateau s_{mid} at the middle-layer boundary, and holds at s_{mid} through the remaining layers.

As a preliminary evaluation, we compare the perplexity of the layer-wise schedule against that of uniform sparsity at equivalent average sparsity on gemma-2-2b, using both WANDA and SPARSEGPT. Perplexity serves as a coarse measure to evaluate the model as a whole. As seen in Table 3, the layer-wise schedule yields lower perplexity in both cases, suggesting that the layer sensitivity profile is a practically useful signal for compression scheduling. We view this as an encouraging preliminary result rather than a fully developed method.

Non-uniform layerwise sparsity has previously been explored for general LLM compression. OWL (Yin et al., 2025) allocates sparsity in proportion to each layer’s activation-outlier ratio, validated against perplexity and downstream task accuracy. Our layer-wise schedule is motivated by a different, interpretability-specific diagnostic. The two criteria are not guaranteed to coincide: outlier concentration and SAE-degradation sensitivity are distinct properties of a layer’s activation distribution, though a systematic comparison is orthogonal to this paper’s central contribution, and we leave it to future work.

Cross-model generalisation confirms the theoretical ordering. Table 2 reports results aggregated over all layers for four model architectures spanning roughly two orders of magnitude in parameter count: pythia-70m, gemma-2-2b, gemma-2-9b, and mistral-7b. Across all models for which valid SAE evaluations are available, the method ordering SPARSEGPT $\succ$ WANDA $\succ$ MAGNITUDE is preserved on Core metrics, with WANDA and

Model	Avg. Sparsity	Wanda		SparseGPT	
		Uniform	Layer-wise	Uniform	Layer-wise
gemma-2-2b	50%	212	148	141	98

Table 3: Perplexity on gemma-2-2b under uniform versus layer-wise sparsity allocation evaluated for WANDA and SPARSEGPT. The ramp schedule assigns 20% sparsity to early layers, increasing linearly to 60% at the middle-layer and holding at 60% until the last layer.

SPARSEGPT achieving near-baseline KL and CE scores while MAGNITUDE pruning causes marked degradation. The result is especially pronounced for pythia-70m, where MAGNITUDE pruning reduces the KL divergence score from 0.926 to 0.642, a collapse not observed under WANDA (0.880) or SPARSEGPT (0.882). Mistral-7b also displays comparatively mild absolute degradation across all three methods on Core metrics.

7 Conclusion

This paper studies the interaction between weight pruning and mechanistic interpretability via sparse autoencoders (SAE), showing that, for a fixed SAE, pruning-induced SAE degradation is governed by perturbation energy, a covariance-weighted norm that measures how strongly pruning distorts the activation distribution, modulated by the SAE’s Lipschitz constant. This framework explains the consistent empirical ordering of pruning methods: MAGNITUDE pruning, which ignores activation geometry, induces large perturbation energy and significantly degrades SAE behavior, while activation-aware methods such as WANDA and SPARSEGPT better preserve interpretability by controlling perturbation energy. We validate this theory through a comprehensive evaluation across gemma-2-2b, gemma-2-9b, mistral-7b, and pythia-70m, using pretrained SAEs and the SAE Bench evaluation suite. Additionally, we identify a structural vulnerability in middle layers and show that layer-wise sparsity allocation, protecting these layers better, preserves representations under equivalent compression.

Limitations and Future Work. Our study has several limitations that point toward promising directions for future work. **First**, we hold SAE weights fixed after pruning. An important complementary direction is to retrain SAEs from scratch on the pruned model’s activations and examine whether the recovered features remain consistent with those of the dense model. **Second**, our conclusions rest on four model–SAE pairs drawn from a relatively homogeneous set of architectures. It remains unclear whether the results generalize to other model families, instruction-tuned models, mixture-of-experts architectures, or SAEs trained on MLP and attention-head outputs rather than the residual stream. **Third**, our analysis is confined to unstructured pruning; developing analogous theory for structured pruning, quantization, and knowledge distillation is important as these methods grow more prevalent in production deployments. **Fourth**, our layer-wise sparsity allocation is a preliminary illustration that the sensitivity profile is a useful scheduling signal, not a proposed allocation method in its own right: we do not benchmark it against more non-intuitive methods or dedicated layerwise-allocation methods such as OWL Yin et al. (2025), since a rigorous comparison is a separate undertaking from the SAE-degradation analysis that is this paper’s focus. **Finally**, while the middle-layer sensitivity finding is empirically robust, our framework does not yet explain it theoretically. A formal account of how the structure of internal representations evolves with depth, and how this motivates principled rather than heuristic sparsity allocation, remains an open problem.

Acknowledgments

This work was funded in part by the National Science Foundation under award number IIS2202699, IIS-2416895, IIS-2301599, CMMI2301601, DMS-2529302. We used large language models (Claude by Anthropic, ChatGPT by OpenAI) to assist with writing and paraphrasing portions of this paper. No LLMs were used to originate research ideas, generate data or plots, or perform evaluation.

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A Appendix

A.1 Detailed Proofs for the Perturbation-Theoretic Framework

This appendix provides fully detailed, self-contained proofs of the perturbation theory presented in Section 3.

A.1.1 Setup and Notation

We recall the objects involved.

- $W \in \mathbb{R}^{d \times p}$: the weight matrix of a single linear layer.
- $x_{\text{in}} \in \mathbb{R}^p$: the random input to that layer, drawn from the data distribution $\mathcal{D}$.
- $x = Wx_{\text{in}} \in \mathbb{R}^d$: the original (pre-pruning) activation.

- $\Delta W \in \mathbb{R}^{d \times p}$: the weight *change* induced by pruning. Because pruning zeroes entries, every non-zero entry (i, j) of ΔW satisfies $(\Delta W)_{ij} = -W_{ij}$.
- $x' = (W + \Delta W)x_{\text{in}} = x + \delta$: the pruned activation, where $\delta := \Delta W x_{\text{in}}$ is the **perturbation vector**.
- $f_\theta : \mathbb{R}^d \rightarrow \mathbb{R}^d$: the trained SAE, which is L -Lipschitz (Assumption 2.2).
- $\Sigma_{x_{\text{in}}} := \mathbb{E}[x_{\text{in}} x_{\text{in}}^\top] \in \mathbb{R}^{p \times p}$: the second-moment (input covariance) matrix.

We want to bound the **expected squared reconstruction error on pruned activations**:

$$\mathcal{E}_{\text{pruned}} := \mathbb{E} \left[\|x' - f_\theta(x')\|^2 \right]. \quad (10)$$

We will show that this is bounded in terms of the **perturbation energy**

$$\varepsilon^2 := \mathbb{E} \left[\|\delta\|^2 \right], \quad (11)$$

and then show that $\varepsilon^2 = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)$.

A.1.2 Proof of Lemma 3.1

Lemma (Lemma 3.1, restated). Let f_θ be L -Lipschitz and let $x' = x + \delta$. Then:

$$\|x' - f_\theta(x')\| \leq \|x - f_\theta(x)\| + (1 + L)\|\delta\|. \quad (12)$$

Proof. We want to bound the reconstruction error at x' . We will relate it to the reconstruction error at x , which we know is small, and the perturbation δ , which we want to bound.

Applying the triangle inequality: The triangle inequality states that for any three vectors a, b, c : $\|a - c\| \leq \|a - b\| + \|b - c\|$. We use it with $a = x'$, $c = f_\theta(x')$, and b chosen to introduce $f_\theta(x)$:

$$\begin{aligned} \|x' - f_\theta(x')\| &= \|(x' - f_\theta(x)) + (f_\theta(x) - f_\theta(x'))\| \\ &\leq \|x' - f_\theta(x)\| + \|f_\theta(x) - f_\theta(x')\|. \end{aligned} \quad (13)$$

We now bound each of these two terms separately.

Bounding the first term: Since $x' = x + \delta$, we apply the triangle inequality again:

$$\begin{aligned} \|x' - f_\theta(x)\| &= \|(x + \delta) - f_\theta(x)\| \\ &= \|(x - f_\theta(x)) + \delta\| \\ &\leq \|x - f_\theta(x)\| + \|\delta\|. \end{aligned} \quad (14)$$

Bounding the second term: We use the Lipschitz continuity assumption here. By Assumption 2.2, the SAE satisfies $\|f_\theta(u) - f_\theta(v)\| \leq L\|u - v\|$ for all u, v . Applying this with $u = x$ and $v = x' = x + \delta$:

$$\|f_\theta(x) - f_\theta(x')\| \leq L\|x - x'\| = L\|(x) - (x + \delta)\| = L\|\delta\|. \quad (15)$$

Substituting (14) and (15) into (13):

$$\begin{aligned} \|x' - f_\theta(x')\| &\leq \|x' - f_\theta(x)\| + \|f_\theta(x) - f_\theta(x')\| \\ &\leq (\|x - f_\theta(x)\| + \|\delta\|) + L\|\delta\| \\ &= \|x - f_\theta(x)\| + (1 + L)\|\delta\|. \end{aligned} \quad \square$$

Remark (Interpretation of $(1 + L)$). The factor $(1 + L)$ has a precise meaning. The $+1$ accounts for the direct shift of the activation: moving from x to $x' = x + \delta$ shifts the reconstruction target by $\|\delta\|$. The $+L$ accounts for the SAE's *response* to that shift: because the SAE is Lipschitz with constant L , its output can move by at most $L\|\delta\|$ when the input moves by $\|\delta\|$. Both movements contribute additively to the reconstruction error.

Proof of Theorem 3.2 (Expected Bound and Covariance Decomposition)

Theorem (Theorem 3.2, restated). Under Assumption 2.2:

$$\mathbb{E} \left[\|x' - f_\theta(x')\|^2 \right] \leq \mathbb{E} \left[\|x - f_\theta(x)\|^2 \right] + (1+L)^2 \varepsilon^2 + 2(1+L) \sqrt{\mathbb{E}[\|x - f_\theta(x)\|^2] \cdot \varepsilon^2}, \quad (16)$$

where

$$\varepsilon^2 := \mathbb{E} \left[\|\delta\|^2 \right] = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top). \quad (17)$$

Proof. We prove the two parts separately.

Part 1: Proving the bound (16).

From Lemma 3.1, we have for every sample:

$$\|x' - f_\theta(x')\| \leq \|x - f_\theta(x)\| + (1+L)\|\delta\|. \quad (18)$$

Let us write $A := \|x - f_\theta(x)\|$ and $B := (1+L)\|\delta\|$ to keep the algebra clean. The bound says $\|x' - f_\theta(x')\| \leq A + B$, so since squaring is monotone on non-negative numbers:

$$\|x' - f_\theta(x')\|^2 \leq (A+B)^2 = A^2 + 2AB + B^2. \quad (19)$$

Expanding back:

$$\|x' - f_\theta(x')\|^2 \leq \|x - f_\theta(x)\|^2 + 2(1+L)\|x - f_\theta(x)\| \cdot \|\delta\| + (1+L)^2 \|\delta\|^2. \quad (20)$$

$\mathbb{E}[\cdot]$ is linear, so taking expectations on both sides of (20):

$$\begin{aligned} \mathbb{E} \left[\|x' - f_\theta(x')\|^2 \right] &\leq \mathbb{E} \left[\|x - f_\theta(x)\|^2 \right] + (1+L)^2 \mathbb{E} \left[\|\delta\|^2 \right] \\ &\quad + 2(1+L) \mathbb{E} \left[\|x - f_\theta(x)\| \cdot \|\delta\| \right]. \end{aligned} \quad (21)$$

The cross term $\mathbb{E}[\|x - f_\theta(x)\| \cdot \|\delta\|]$ involves the expectation of a *product* of two random variables. We bound it using the Cauchy–Schwarz inequality for expectations, which states that for any two random variables U, V :

$$\mathbb{E}[UV] \leq \sqrt{\mathbb{E}[U^2] \mathbb{E}[V^2]}. \quad (22)$$

Setting $U = \|x - f_\theta(x)\|$ and $V = \|\delta\|$:

$$\mathbb{E} \left[\|x - f_\theta(x)\| \cdot \|\delta\| \right] \leq \sqrt{\mathbb{E} \left[\|x - f_\theta(x)\|^2 \right] \cdot \mathbb{E} \left[\|\delta\|^2 \right]}. \quad (23)$$

Substituting (23) into (21) and writing $\varepsilon^2 := \mathbb{E}[\|\delta\|^2]$:

$$\mathbb{E} \left[\|x' - f_\theta(x')\|^2 \right] \leq \mathbb{E} \left[\|x - f_\theta(x)\|^2 \right] + (1+L)^2 \varepsilon^2 + 2(1+L) \sqrt{\mathbb{E} \left[\|x - f_\theta(x)\|^2 \right] \cdot \varepsilon^2}. \quad (24)$$

This is precisely (16). $\square_{\text{Part 1}}$

Part 2: Proving the covariance identity (17).

We show that $\mathbb{E}[\|\delta\|^2] = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)$.

Recall $\delta = \Delta W x_{\text{in}}$. The squared norm of a vector v can be written as the dot product $\|v\|^2 = v^\top v$. Therefore:

$$\|\delta\|^2 = \delta^\top \delta = (\Delta W x_{\text{in}})^\top (\Delta W x_{\text{in}}) = x_{\text{in}}^\top \Delta W^\top \Delta W x_{\text{in}}. \quad (25)$$

For any deterministic matrix $M \in \mathbb{R}^{p \times p}$ and any random vector $v \in \mathbb{R}^p$, following is a fundamental identity:

$$\mathbb{E} \left[v^\top M v \right] = \text{tr}(M \mathbb{E}[v v^\top]). \quad (26)$$

Why does this identity hold? Writing out the quadratic form entry-by-entry: $v^\top M v = \sum_{i,j} M_{ij} v_i v_j$. Taking expectations and using linearity: $\mathbb{E}[v^\top M v] = \sum_{i,j} M_{ij} \mathbb{E}[v_i v_j]$. But $\mathbb{E}[v_i v_j] = [\mathbb{E}[v v^\top]]_{ij}$, so:

$$\mathbb{E}[v^\top M v] = \sum_{i,j} M_{ij} [\mathbb{E}[v v^\top]]_{ij} = \sum_i [M \mathbb{E}[v v^\top]]_{ii} = \text{tr}(M \mathbb{E}[v v^\top]), \quad (27)$$

where the second equality uses the definition of matrix multiplication, and the third uses the definition of the trace of a matrix.

Applying (26) with $M = \Delta W^\top \Delta W$ and $v = x_{\text{in}}$:

$$\mathbb{E}[\|\delta\|^2] = \mathbb{E}[x_{\text{in}}^\top \Delta W^\top \Delta W x_{\text{in}}] = \text{tr}(\Delta W^\top \Delta W \mathbb{E}[x_{\text{in}} x_{\text{in}}^\top]) = \text{tr}(\Delta W^\top \Delta W \Sigma_{x_{\text{in}}}). \quad (28)$$

The trace satisfies $\text{tr}(AB) = \text{tr}(BA)$ for any matrices A, B of compatible dimensions (according to the *cyclic property*). Applying this with $A = \Delta W^\top$ and $B = \Delta W \Sigma_{x_{\text{in}}}$:

$$\text{tr}(\Delta W^\top \Delta W \Sigma_{x_{\text{in}}}) = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top). \quad (29)$$

Chaining the equalities :

$$\varepsilon^2 = \mathbb{E}[\|\delta\|^2] = \mathbb{E}[\|\Delta W x_{\text{in}}\|^2] = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top). \quad (30)$$

□

Remark (Eigenvalue expansion of ε^2). Writing the eigendecomposition $\Sigma_{x_{\text{in}}} = \sum_{i=1}^p \lambda_i v_i v_i^\top$ with $\lambda_1 \geq \dots \geq \lambda_p > 0$, the cyclic property gives:

$$\varepsilon^2 = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top) = \sum_{i=1}^p \lambda_i \|\Delta W v_i\|^2. \quad (31)$$

Each term $\lambda_i \|\Delta W v_i\|^2$ is the product of (i) the variance of the data in direction v_i , and (ii) the squared magnitude of the activation perturbation in that direction. Equation (31) makes the covariance-weighting explicit: a pruning decision that perturbs a high-variance direction (large λ_i) is penalised proportionally more in ε^2 , and therefore causes proportionally more damage to the SAE.

A.1.3 Proof of Corollary 3.3 (Paradigm Comparison)

Corollary (Corollary 3.3, restated). At any target sparsity s , the three pruning methods relate to ε^2 as follows:

1. MAGNITUDE pruning minimises $\|\Delta W\|_F^2 = \text{tr}(\Delta W I \Delta W^\top)$.
2. WANDA minimises $\text{tr}(\Delta W \text{diag}(\mathbb{E}[x_{\text{in}} x_{\text{in}}^\top]) \Delta W^\top)$.
3. SparseGPT approximately minimises $\varepsilon^2 = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)$ directly.

Proof. Each claim follows directly from the definition of the method (Definition 2.3) combined with the identity $\varepsilon^2 = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)$ from Theorem 3.2.

Claim 1 (MAGNITUDE pruning).

MAGNITUDE pruning selects the pruning mask to minimise $\sum_{(i,j) \in S} w_{ij}^2 = \|\Delta W\|_F^2$ (since $(\Delta W)_{ij} = -w_{ij}$ for pruned entries). By the covariance identity, $\|\Delta W\|_F^2 = \text{tr}(\Delta W I \Delta W^\top)$, which equals $\text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)$ only when $\Sigma_{x_{\text{in}}} = I$. When $\Sigma_{x_{\text{in}}} \neq I$ (the typical case), MAGNITUDE pruning minimises the wrong objective.

Claim 2 (WANDA).

WANDA scores each entry (i, j) by $w_{ij}^2 \cdot \mathbb{E}[(x_{\text{in}})_j^2]$ and prunes the lowest-scoring entries. The summed score of all pruned entries is $\sum_{(i,j) \in S} w_{ij}^2 \cdot \mathbb{E}[(x_{\text{in}})_j^2] = \text{tr}(\Delta W D \Delta W^\top)$, where $D = \text{diag}(\mathbb{E}[(x_{\text{in}})^2])$ is the diagonal matrix of per-coordinate input variances. Since $D_{jj} = \mathbb{E}[(x_{\text{in}})_j^2]$, and noting that $[\Sigma_{x_{\text{in}}}]_{jj} = \mathbb{E}[(x_{\text{in}})_j^2]$, the WANDA objective is the *diagonal approximation* to $\text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)$, obtained by discarding the off-diagonal (cross-covariance) terms of $\Sigma_{x_{\text{in}}}$.

Claim 3 (SparseGPT).

SparseGPT solves the layer-wise reconstruction problem $\min_{\Delta W: \text{sparsity}(W+\Delta W)=s} \mathbb{E}[\|\Delta W x_{\text{in}}\|^2]$ via a second-order approximation using the empirical Hessian $\hat{\Sigma}_{x_{\text{in}}} = \frac{1}{N} \sum_{n=1}^N x_{\text{in}}^{(n)} (x_{\text{in}}^{(n)})^\top$. By Theorem 3.2, $\mathbb{E}[\|\Delta W x_{\text{in}}\|^2] = \text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)$, so SparseGPT directly (approximately) minimises ε^2 . $\square$

Remark A.1 (Hierarchy of approximations). The three methods form a natural hierarchy in how faithfully they approximate the true objective ε^2 :

$$\underbrace{\text{tr}(\Delta W I \Delta W^\top)}_{\text{MAGNITUDE}} \prec \underbrace{\text{tr}(\Delta W \text{diag}(\Sigma_{x_{\text{in}}}) \Delta W^\top)}_{\text{WANDA}} \prec \underbrace{\text{tr}(\Delta W \Sigma_{x_{\text{in}}} \Delta W^\top)}_{\text{SparseGPT}}, \quad (32)$$

where $\prec$ denotes “is a coarser approximation of the true perturbation energy than”. MAGNITUDE pruning ignores $\Sigma_{x_{\text{in}}}$ entirely; WANDA uses only its diagonal; SparseGPT uses the full matrix.

A.2 SAEBench Evaluation Suite

This appendix gives a detailed overview of the SAEBench evaluation suite (Karvonen et al., 2025). We also provide intuitive understanding of each metric category (Chanin et al., 2025; Karvonen et al., 2024; 2025). Ultimately, we conclude the appendix by talking about the rationale behind the SCR and TPP threshold choices.

A.2.1 Core metrics

KL divergence score. Let $\mathcal{D}_{\text{orig}}$ denote the output distribution of the original model and $\mathcal{D}_{\text{SAE}}$ the output distribution when the residual stream is replaced by the SAE reconstruction. The zero-ablation distribution $\mathcal{D}_{\text{zero}}$ is obtained by replacing the residual stream with zeros. The KL divergence score is defined as:

$$\text{KL score} = 1 - \frac{D_{\text{KL}}(\mathcal{D}_{\text{orig}} \parallel \mathcal{D}_{\text{SAE}})}{D_{\text{KL}}(\mathcal{D}_{\text{orig}} \parallel \mathcal{D}_{\text{zero}})}, \quad (33)$$

normalized so that a perfect reconstruction yields a score of 1 and a zero-ablation yields a score of 0. This metric is sensitive to distributional shifts in the model’s predictions, not just pointwise reconstruction error.

CE loss score. Let H_{orig} , H^* , and H_{ablation} denote the cross-entropy loss of the model under the original activations, SAE-reconstructed activations, and zero-ablated activations, respectively. The CE loss score is defined as:

$$\text{CE score} = \frac{H_{\text{ablation}} - H^*}{H_{\text{ablation}} - H_{\text{orig}}}. \quad (34)$$

A score of 1 indicates that the SAE reconstruction perfectly recovers the original model’s cross-entropy loss, while a score of 0 corresponds to the zero-ablation baseline. Unlike the KL divergence score, this metric emphasizes the impact of reconstruction quality on next-token prediction performance.

Explained variance. For a layer with activation $x \in \mathbb{R}^d$ and SAE reconstruction $\hat{x} = f_\theta(x)$, the explained variance is:

$$\text{EV} = 1 - \frac{\text{Var}(x - \hat{x})}{\text{Var}(x)}, \quad (35)$$

where variance is computed over the token dimension. A higher explained variance indicates that the SAE accounts for more of the variance in the original activations. This metric is sensitive to systematic reconstruction failures that affect large portions of the activation space.

Mean squared error (MSE). The MSE between the original activation x and its reconstruction $\hat{x}$ is:

$$\text{MSE} = \mathbb{E} \left[\|x - f_\theta(x)\|^2 \right], \quad (36)$$

measured in the same units as the activation space. Unlike the normalized metrics above, MSE is an absolute measure and therefore varies substantially across layers and model sizes, as later layers tend to have activations with larger norms. We report MSE alongside the normalized metrics to provide a scale-sensitive view of reconstruction quality.

Cosine similarity. The mean cosine similarity between original and reconstructed activations is:

$$\text{Cos Sim} = \mathbb{E} \left[\frac{\langle x, f_\theta(x) \rangle}{\|x\| \|f_\theta(x)\|} \right]. \quad (37)$$

This metric captures directional alignment between original and reconstructed activations, independent of their magnitudes. It is particularly informative when activation norms vary substantially across layers, since magnitude-preserving but direction-distorting perturbations appear benign under MSE yet are detected here.

A.2.2 Feature Absorption

Feature absorption happens when SAEs, due to sparsity pressure, fail to represent a general feature directly. Instead of having both a general latent (e.g., “starts with S”) and specific latents (e.g., “short”), the SAE only activates the specific ones. This makes the general latent look unreliable $\implies$ The general latent may fire for many tokens but mysteriously may miss some cases, because those cases were absorbed into other latents.

SAEBench Utilization

- Picking a ground-truth concept (e.g., “starts with S”).
- Identifying the SAE latent most responsible for that concept.
- Checking where that latent fails to fire.
- Seeing if other latents step in to cover the missing cases.

The final absorption score is the fraction of a concept’s cases that are not covered by its main latent but are covered by other latents.

Intuitive Example Suppose the set of all S-words is:

short, small, sand, sun, sea, soap, star, slow, sound, sharp

SAEBench finds that latent L_S is the best candidate for the concept starts with S. Let us assume that L_S fires on 7 out of 10 tokens: sand, sun, sea, soap, star, slow, sharp, and misses the tokens: short, small, sound. Thus, the recall of the latent $L_S = \frac{7}{10} = 70\%$.

SAEBench then checks whether other latents cover the missing three tokens. Let us assume the latent L_{short} fires on tokens short and small, and L_{sound} fires on the token sound. Now all missing cases are covered, but by different latents.

Raw absorption = 30% (because 3/10 cases were absorbed away from the latent L_5). Each missing case is explained elsewhere, so they are not pure false negatives. So the score reported would be 30% absorption, with full compensation of the missing cases by other latents. If none of the missing tokens had been covered elsewhere, the absorption score would flag this as a bad collapse.

Thus, the absorption score in SAEBench = “How often is a general feature incomplete, and to what extent do other latents pick up the missing pieces?”

Lower absorption scores → better interpretability! A low absorption score means the general latent is reliable. It means this latent fires consistently when the concept is present in the input. A tiny portion of any concept is fragmented into other specific latents.

Sparsity(L_0) defines the number of latents that are allowed to fire per token. Low L_0 (high sparsity) implies very few latents fire, high L_0 (low sparsity) implies many latents can fire. From the SAEBench results, we can infer that at low L_0 , SAEs aggressively minimize number of active latents $\Rightarrow$ high absorption score.

A.2.3 Spurious Correlation Coefficient

SCR score quantifies the capability of an SAE to find and remove spurious correlations (irrelevant signals) while preserving the signal we actually care about.

SAEBench Utilization SAEBench takes biased datasets. It uses probes and SAE latents to identify the spurious features (bias-creating features). Then it ablates those latents and checks if classifier accuracy on the actual task improves.

Intuitive Example Suppose we have a dataset with biased representation, featuring male professors and female nurses. The task is to predict a person’s profession based on their individual characteristics. Here, gender is the spurious feature.

We train a classifier on this biased data, which will learn the gender–profession shortcut. Let us assume:

$$\text{Accuracy on biased data}(A_{base}) = 95\%$$

Next, we include female professors and male nurses in the dataset, and again train a classifier. Let us assume:

$$\text{Accuracy on unbiased data}(A_{best}) = 75\%$$

Ultimately, we use SAE latents to ablate spurious features. We find gender-related latents, set them to zero, and retrain the classifier on the biased data. Let us assume:

$$\text{Accuracy on biased data with ablated latents}(A_{abl}) = 80\%$$

Finally, we compute the SCR score as follows:

$$\begin{aligned} SCR &= \frac{A_{abl} - A_{base}}{A_{best} - A_{base}} \\ &= \frac{80 - 95}{75 - 95} = 0.75 \end{aligned}$$

Higher SCR scores → better SAE! A score of 1.0 indicates that SAE removed all spurious bias perfectly. Thus, SCR tells you whether an SAE can cut out spurious shortcuts like gender bias while keeping the real task intact.

A.2.4 Targeted Probe Perturbation

TPP score quantifies the capability of an SAE to isolate individual concepts so that removing its latents only affects that concept and leaves others unchanged.

SAEBench Utilization SAEBench applies TPP to any multiclass dataset to test whether SAE features for one class are cleanly separated from others. We train probes (small classifiers) for each class. Next, we find the SAE latents most used by a specific probe and ablate them.

TPP is a generalization of SCR. Instead of needing biased datasets, it can test disentanglement on any multiclass dataset.

Intuitive Example Suppose we have an Amazon reviews dataset with books, electronics, home supplies and the goal is to check if book latents are isolated.

We train probes for each class. Let us assume:

$$\begin{aligned} \text{Accuracy for Books probe} &= 85\% \\ \text{Accuracy for Electronics probe} &= 75\% \\ \text{Accuracy for Home supplies probe} &= 80\% \end{aligned}$$

After ablating book probes, let us assume:

$$\begin{aligned} \text{Accuracy for Books probe} &= 55\% \implies \Delta = -30\% \\ \text{Accuracy for Electronics probe} &= 70\% \implies \Delta = -5\% \\ \text{Accuracy for Home supplies probe} &= 79\% \implies \Delta = -1\% \end{aligned}$$

Since we are playing around with the target: books, we compute the TPP scores as follows:

$$\begin{aligned} \text{TPP} &= \text{mean}\Delta_{i=j} - \text{mean}\Delta_{i \neq j} \\ &= (-30) - \left(\frac{-5 - 1}{2}\right) \\ &= -30 + 3 = -27 \end{aligned}$$

Since TPP is normalized in practice, this would be reported as a high positive score showing the Book latents only hurt Book classification.

Higher TPP scores $\rightarrow$ better SAE! A higher TPP score is better, indicating that the SAE identified features specific to one class. Removing them tanks the target probe but leaves others intact. A lower TPP score means SAE latents are entangled.

A.2.5 Threshold Sensitivity of SCR and TPP

SCR and TPP are each computed at seven ablation thresholds $k \in \{2, 5, 10, 20, 50, 100, 500\}$, corresponding to the number of top-scoring SAE latents ablated per concept direction. The choice of k is a hyperparameter of the evaluation, not of the model or the pruning method, and a valid evaluation protocol should not depend sensitively on it.

We verify this for gemma-2-2b at 50% sparsity in Figure 4, averaging across all layers. At every value of k the ordering $\text{SPARSEGPT} \geq \text{WANDA} \geq \text{MAGNITUDE}$ holds for both SCR and TPP.

Figure 4 examines how SCR and TPP vary with the ablation threshold k . The raw metric values for all methods increase with k for both SCR and TPP, as ablating more latents captures more of the relevant concept signal. The method ordering is consistent across all evaluated thresholds: MAGNITUDE pruning shows the largest degradation throughout, while WANDA and SPARSEGPT remain close to each other and substantially closer to the dense baseline. The percentage change from the dense baseline stabilises for $k \geq 10$ for the activation-aware methods, while MAGNITUDE pruning shows larger and less stable degradation across thresholds. Based on these observations, we can confirm the choice of $k = 10$ as the canonical threshold, as it lies in the stable regime and produces the clearest separation between pruning paradigms.

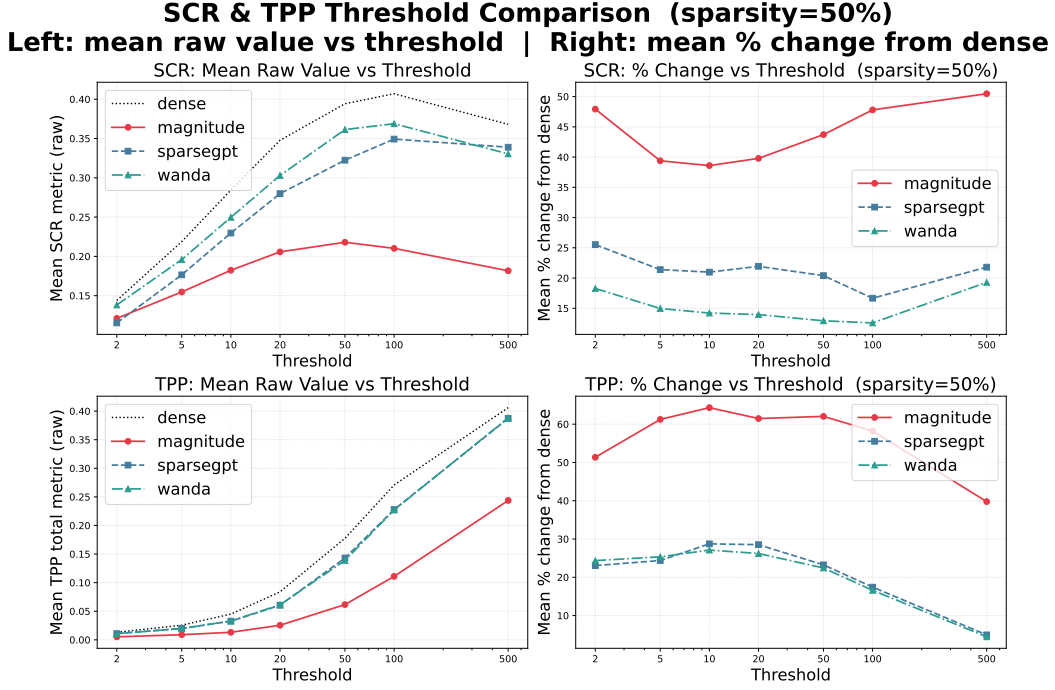


Figure 4: SCR/TPP metric stability across ablation thresholds for gemma-2-2b at 50% sparsity. **Left column:** mean raw metric values as a function of ablation threshold $k \in \{2, 5, 10, 20, 50, 100, 500\}$ for SCR (top)/ TPP (bottom), alongside the dense baseline (dotted). **Right column:** corresponding mean percentage change from the dense baseline.

A.3 Per-Layer Results for gemma-2-2b

Full layer-by-layer SAE Bench metrics for gemma-2-2b at 50% sparsity, split by depth group. These results support the aggregated group-level summaries reported in Table 1 and the per-layer profiles shown in Figures 1 and 2.

A.4 Empirical Perturbation Energy

Table 7 reports the empirical perturbation energy $\varepsilon^2 = \mathbb{E}[\|\Delta W x_{\text{in}}\|^2]$, averaged within each layer group, for each pruning method on gemma-2-2b at 50% sparsity, estimated over the calibration set of Sec. 5. The ordering $\varepsilon^2(\text{SPARSEGPT}) < \varepsilon^2(\text{WANDA}) < \varepsilon^2(\text{MAGNITUDE})$ predicted by Corollary 3.3 is confirmed at every layer group: MAGNITUDE incurs roughly twice the perturbation energy of the activation-aware methods. This moderate ε^2 ratio is amplified into the large SAE Bench gaps of Table 1 through the quadratic and cross terms of Theorem 3.2, consistent with the bound’s structure.

Notably, per-layer ε^2 does not mirror the layer sensitivity profile of Figure 3: middle layers exhibit the lowest local ε^2 yet the highest SAE degradation. Middle-layer vulnerability is therefore driven not by locally larger perturbation but by the accumulation of upstream perturbation from early layers propagating through residual connections. This further motivates the layer-wise allocation of Sec. 6: protecting early layers reduces the perturbation that cascades into vulnerable middle layers.

Layers	Method	Spar. (%)	Core					Absorption		SCR	TPP
			KL Score↑	CE Score↑	Expl. Var.↑	Cos Sim↑	MSE ↓	Abs. Frac.↓	Full Abs.↓	Top-10 ↑	Top-10 ↑
Layer 0	<i>Baseline</i>	0	0.9966	0.9967	0.9453	0.9766	0.1396	0.0131	0.0324	0.1488	0.0272
	MAGNITUDE	50	0.9915	0.9962	0.8984	0.9531	0.2754	0.0141	0.0432	0.1385	0.0327
	WANDA	50	0.9964	0.9932	0.9375	0.9727	0.1582	0.0127	0.0339	0.1605	0.0242
	SPARSEGPT	50	0.9953	0.9948	0.9375	0.9727	0.1611	0.0133	0.0346	0.1514	0.0206
Layer 1	<i>Baseline</i>	0	0.9958	0.9951	0.9219	0.9648	0.2402	0.0989	0.0913	0.1788	0.0336
	MAGNITUDE	50	0.9914	0.9962	0.8750	0.9375	0.3984	0.1130	0.1012	0.1456	0.0357
	WANDA	50	0.9956	0.9914	0.9141	0.9609	0.2793	0.0967	0.0820	0.1732	0.0241
	SPARSEGPT	50	0.9942	0.9931	0.9102	0.9570	0.2793	0.0853	0.0807	0.1602	0.0271
Layer 2	<i>Baseline</i>	0	0.9960	0.9967	0.9141	0.9609	0.2578	0.0900	0.0224	0.1279	0.0184
	MAGNITUDE	50	0.9912	0.9962	0.8516	0.9219	0.4512	0.1002	0.0458	0.1024	0.0195
	WANDA	50	0.9956	0.9914	0.8945	0.9492	0.3008	0.0880	0.0246	0.1048	0.0105
	SPARSEGPT	50	0.9943	0.9948	0.8867	0.9453	0.3086	0.0748	0.0226	0.0870	0.0120
Layer 3	<i>Baseline</i>	0	0.9930	0.9934	0.8828	0.9453	0.3906	0.1136	0.0570	0.1250	0.0190
	MAGNITUDE	50	0.9843	0.9772	0.8125	0.9023	0.6211	0.1027	0.0680	0.0971	0.0108
	WANDA	50	0.9912	0.9829	0.8594	0.9336	0.4297	0.1145	0.0614	0.1203	0.0142
	SPARSEGPT	50	0.9899	0.9880	0.8516	0.9258	0.4512	0.1182	0.0704	0.1202	0.0153
Layer 4	<i>Baseline</i>	0	0.9964	0.9967	0.9062	0.9531	0.3320	0.1330	0.0387	0.1160	0.0278
	MAGNITUDE	50	0.9876	0.9848	0.8398	0.9180	0.5352	0.1736	0.0736	0.0824	0.0144
	WANDA	50	0.9957	0.9914	0.8828	0.9414	0.3711	0.1365	0.0319	0.1104	0.0154
	SPARSEGPT	50	0.9939	0.9931	0.8789	0.9375	0.3848	0.1657	0.0552	0.0951	0.0155
Layer 5	<i>Baseline</i>	0	0.9930	0.9918	0.8711	0.9336	0.5312	0.0444	0.0212	0.1551	0.0159
	MAGNITUDE	50	0.9752	0.9582	0.8203	0.9062	0.7812	0.0543	0.0316	0.1002	0.0194
	WANDA	50	0.9916	0.9846	0.8516	0.9258	0.5547	0.0492	0.0261	0.1303	0.0182
	SPARSEGPT	50	0.9888	0.9863	0.8477	0.9219	0.5859	0.0570	0.0302	0.1232	0.0182
Layer 6	<i>Baseline</i>	0	0.9937	0.9934	0.8750	0.9375	0.5352	0.0393	0.0134	0.1455	0.0334
	MAGNITUDE	50	0.9817	0.9772	0.8203	0.9062	0.7656	0.0504	0.0252	0.0988	0.0226
	WANDA	50	0.9923	0.9846	0.8594	0.9297	0.5742	0.0454	0.0189	0.1360	0.0245
	SPARSEGPT	50	0.9906	0.9897	0.8516	0.9258	0.5977	0.0454	0.0217	0.1189	0.0288
Layer 7	<i>Baseline</i>	0	0.9928	0.9918	0.8828	0.9375	0.6602	0.0347	0.0164	0.2941	0.0324
	MAGNITUDE	50	0.9794	0.9734	0.8281	0.9062	0.9688	0.0471	0.0259	0.1569	0.0084
	WANDA	50	0.9913	0.9829	0.8594	0.9258	0.7109	0.0475	0.0201	0.2210	0.0173
	SPARSEGPT	50	0.9899	0.9880	0.8477	0.9180	0.7656	0.0508	0.0222	0.2015	0.0244
Layer 8	<i>Baseline</i>	0	0.9925	0.9918	0.8789	0.9375	0.7539	0.0395	0.0255	0.2509	0.0354
	MAGNITUDE	50	0.9797	0.9772	0.8281	0.9062	1.0859	0.0323	0.0231	0.0951	0.0049
	WANDA	50	0.9914	0.9829	0.8555	0.9219	0.8047	0.0314	0.0192	0.1824	0.0207
	SPARSEGPT	50	0.9898	0.9880	0.8438	0.9141	0.8633	0.0352	0.0245	0.1383	0.0142

Table 4: Per-layer SAEbench metrics for gemma-2-2b at 50% sparsity, layers 0–8 (Early group). The dense baseline is italicised for each layer. ↑ higher is better; ↓ lower is better.

Layers	Method	Spar. (%)	Core					Absorption		SCR	TPP
			KL Score↑	CE Score↑	Expl. Var.↑	Cos Sim↑	MSE ↓	Abs. Frac.↓	Full Abs.↓	Top-10 ↑	Top-10 ↑
Layer 9	<i>Baseline</i>	0	0.9917	0.9901	0.8711	0.9297	0.9844	0.0555	0.0498	0.2386	0.0247
	MAGNITUDE	50	0.9664	0.9544	0.7773	0.8789	1.3438	0.0555	0.0498	0.0858	0.0118
	WANDA	50	0.9915	0.9846	0.8438	0.9141	1.0234	0.0429	0.0310	0.1830	0.0170
	SPARSEGPT	50	0.9888	0.9880	0.8281	0.9062	1.0938	0.0527	0.0414	0.1565	0.0138
Layer 10	<i>Baseline</i>	0	0.9909	0.9901	0.8672	0.9258	1.1953	0.0708	0.0541	0.2430	0.0220
	MAGNITUDE	50	0.9733	0.9620	0.7812	0.8789	1.6719	0.0967	0.0769	0.0921	0.0081
	WANDA	50	0.9914	0.9863	0.8438	0.9141	1.2109	0.0701	0.0518	0.1949	0.0153
	SPARSEGPT	50	0.9887	0.9897	0.8203	0.9062	1.3281	0.0834	0.0659	0.1593	0.0159
Layer 11	<i>Baseline</i>	0	0.9900	0.9885	0.8750	0.9258	1.4844	0.1061	0.0916	0.2557	0.0354
	MAGNITUDE	50	0.9649	0.9430	0.7695	0.8711	2.2344	0.1256	0.1183	0.0720	0.0064
	WANDA	50	0.9908	0.9863	0.8516	0.9102	1.5000	0.1020	0.0885	0.2178	0.0265
	SPARSEGPT	50	0.9883	0.9897	0.8203	0.8984	1.7109	0.1181	0.1046	0.1827	0.0184
Layer 12	<i>Baseline</i>	0	0.9896	0.9885	0.8750	0.9219	1.5391	0.0992	0.0840	0.2924	0.0250
	MAGNITUDE	50	0.9736	0.9696	0.7773	0.8750	2.0781	0.1212	0.0971	0.1018	0.0056
	WANDA	50	0.9904	0.9863	0.8516	0.9102	1.5703	0.0987	0.0825	0.2181	0.0242
	SPARSEGPT	50	0.9879	0.9897	0.8164	0.8984	1.8281	0.1112	0.0966	0.2046	0.0206
Layer 13	<i>Baseline</i>	0	0.9885	0.9868	0.8750	0.9297	1.9766	0.1344	0.1252	0.2864	0.0275
	MAGNITUDE	50	0.9692	0.9582	0.7812	0.8828	3.0156	0.1609	0.1517	0.2095	0.0125
	WANDA	50	0.9893	0.9846	0.8477	0.9180	2.1250	0.1279	0.1183	0.2008	0.0318
	SPARSEGPT	50	0.9871	0.9897	0.8242	0.9062	2.5156	0.1422	0.1401	0.1783	0.0152
Layer 14	<i>Baseline</i>	0	0.9885	0.9868	0.8711	0.9297	2.1719	0.0333	0.0288	0.3756	0.0440
	MAGNITUDE	50	0.9562	0.9316	0.7734	0.8750	3.3438	0.0399	0.0390	0.1078	0.0102
	WANDA	50	0.9893	0.9863	0.8438	0.9180	2.3594	0.0354	0.0291	0.2755	0.0439
	SPARSEGPT	50	0.9869	0.9897	0.8281	0.9102	2.7812	0.0401	0.0351	0.2309	0.0380
Layer 15	<i>Baseline</i>	0	0.9872	0.9868	0.8906	0.9375	2.3594	0.0314	0.0274	0.3937	0.0584
	MAGNITUDE	50	0.9700	0.9734	0.8047	0.8945	4.0938	0.0382	0.0390	0.1860	0.0090
	WANDA	50	0.9881	0.9846	0.8594	0.9258	2.6719	0.0308	0.0262	0.2847	0.0378
	SPARSEGPT	50	0.9857	0.9897	0.8398	0.9141	3.1094	0.0377	0.0347	0.2341	0.0356
Layer 16	<i>Baseline</i>	0	0.9862	0.9852	0.8789	0.9336	3.2969	0.0318	0.0254	0.3583	0.0752
	MAGNITUDE	50	0.9566	0.9468	0.7812	0.8789	5.7500	0.0334	0.0319	0.2037	0.0116
	WANDA	50	0.9868	0.9846	0.8555	0.9219	3.6250	0.0283	0.0221	0.3144	0.0377
	SPARSEGPT	50	0.9842	0.9897	0.8320	0.9102	4.2188	0.0317	0.0295	0.3069	0.0398
Layer 17	<i>Baseline</i>	0	0.9859	0.9852	0.8828	0.9375	3.5156	0.0381	0.0409	0.2987	0.0693
	MAGNITUDE	50	0.9496	0.9392	0.7578	0.8672	6.1875	0.0434	0.0538	0.1120	0.0048
	WANDA	50	0.9865	0.9846	0.8594	0.9258	3.9062	0.0581	0.0580	0.2699	0.0389
	SPARSEGPT	50	0.9839	0.9880	0.8398	0.9141	4.4688	0.0410	0.0479	0.2390	0.0351

Table 5: Per-layer SAE Bench metrics for gemma-2-2b at 50% sparsity, layers 9–17 (Middle group). The dense baseline is italicised for each layer. ↑ higher is better; ↓ lower is better.

Layers	Method	Spar. (%)	Core					Absorption		SCR	TPP
			KL Score $\uparrow$	CE Score $\uparrow$	Expl. Var. $\uparrow$	Cos Sim $\uparrow$	MSE $\downarrow$	Abs. Frac. $\downarrow$	Full Abs. $\downarrow$	Top-10 $\uparrow$	Top-10 $\uparrow$
Layer 18	<i>Baseline</i>	0	0.9855	0.9852	0.8867	0.9375	4.4062	0.0951	0.1101	0.3244	0.0655
	MAGNITUDE	50	0.9426	0.9278	0.7617	0.8672	6.9375	0.0795	0.1189	0.1193	0.0063
	WANDA	50	0.9854	0.9846	0.8672	0.9297	4.8750	0.0813	0.0921	0.2761	0.0511
	SPARSEGPT	50	0.9831	0.9880	0.8477	0.9180	5.5000	0.0815	0.1070	0.2658	0.0573
Layer 19	<i>Baseline</i>	0	0.9844	0.9835	0.8828	0.9375	5.5312	0.0823	0.0935	0.3398	0.0660
	MAGNITUDE	50	0.9600	0.9658	0.7773	0.8789	9.3750	0.0787	0.1092	0.1583	0.0126
	WANDA	50	0.9845	0.9846	0.8633	0.9297	6.0312	0.0861	0.0931	0.3061	0.0447
	SPARSEGPT	50	0.9823	0.9880	0.8438	0.9141	6.8750	0.0767	0.0975	0.2888	0.0482
Layer 20	<i>Baseline</i>	0	0.9827	0.9819	0.8750	0.9336	7.2500	0.1144	0.1249	0.4253	0.0677
	MAGNITUDE	50	0.9571	0.9506	0.7617	0.8711	12.5000	0.1091	0.1540	0.2972	0.0093
	WANDA	50	0.9829	0.9829	0.8633	0.9297	7.6562	0.0949	0.1034	0.3849	0.0641
	SPARSEGPT	50	0.9812	0.9863	0.8438	0.9180	8.8125	0.1039	0.1251	0.3876	0.0523
Layer 21	<i>Baseline</i>	0	0.9798	0.9786	0.8750	0.9336	10.0625	0.1283	0.1407	0.4160	0.0846
	MAGNITUDE	50	0.9551	0.9506	0.7891	0.8867	18.7500	0.1043	0.1465	0.4425	0.0144
	WANDA	50	0.9800	0.9795	0.8672	0.9297	10.3750	0.1150	0.1263	0.4694	0.0528
	SPARSEGPT	50	0.9788	0.9828	0.8516	0.9219	12.1250	0.1182	0.1399	0.4662	0.0694
Layer 22	<i>Baseline</i>	0	0.9777	0.9769	0.8633	0.9297	13.9375	0.1217	0.1252	0.4038	0.0619
	MAGNITUDE	50	0.9520	0.9468	0.7578	0.8750	27.2500	0.0751	0.0979	0.4423	0.0034
	WANDA	50	0.9782	0.9777	0.8594	0.9258	14.3750	0.1108	0.1132	0.3853	0.0440
	SPARSEGPT	50	0.9774	0.9828	0.8398	0.9180	17.0000	0.1170	0.1314	0.3808	0.0524
Layer 23	<i>Baseline</i>	0	0.9752	0.9736	0.8555	0.9258	18.3750	0.1120	0.1156	0.3858	0.0634
	MAGNITUDE	50	0.9541	0.9468	0.7422	0.8672	37.5000	0.0672	0.0836	0.4309	0.0098
	WANDA	50	0.9757	0.9743	0.8516	0.9219	18.6250	0.1053	0.1080	0.3862	0.0496
	SPARSEGPT	50	0.9749	0.9794	0.8320	0.9141	22.3750	0.1042	0.1163	0.3495	0.0397
Layer 24	<i>Baseline</i>	0	0.9695	0.9671	0.8438	0.9219	24.8750	0.1410	0.1928	0.3933	0.0638
	MAGNITUDE	50	0.9528	0.9468	0.7266	0.8633	55.0000	0.0747	0.1230	0.3507	0.0136
	WANDA	50	0.9698	0.9675	0.8398	0.9219	25.2500	0.1184	0.1606	0.4123	0.0453
	SPARSEGPT	50	0.9698	0.9725	0.8164	0.9102	30.8750	0.1087	0.1578	0.3605	0.0469
Layer 25	<i>Baseline</i>	0	0.9662	0.9654	0.8633	0.9258	29.8750	0.1717	0.1925	0.4197	0.0722
	MAGNITUDE	50	0.9545	0.9734	0.7188	0.8516	68.5000	0.0761	0.1081	0.3124	0.0240
	WANDA	50	0.9679	0.9675	0.8438	0.9141	30.7500	0.1239	0.1463	0.3742	0.0636
	SPARSEGPT	50	0.9671	0.9725	0.8359	0.9062	37.2500	0.1331	0.1591	0.3850	0.0714

Table 6: Per-layer SAE Bench metrics for gemma-2-2b at 50% sparsity, layers 18–25 (Late group). The dense baseline is italicised for each layer. $\uparrow$ higher is better; $\downarrow$ lower is better.

Layer Group	MAGNITUDE ϵ^2	WANDA ϵ^2	SPARSEGPT ϵ^2
Early (0–8)	57.36	31.87	26.66
Middle (9–17)	47.13	21.16	20.70
Late (18–25)	53.69	26.45	24.95

Table 7: Empirical perturbation energy ϵ^2 by layer group for gemma-2-2b at 50% sparsity. Lower is better; the ordering predicted by Corollary 3.3 holds in every group.